Accepted for publication in *Development Psychology*

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The Impact of Interpersonal Inequality on Children's Prosocial Behavior:

The Critical Role of Perceived Unfairness

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This research was supported by the Eunice Kennedy Shriver National Institute of Child Health & Human Development of the National Institutes of Health, grant R03HD101652 to Amrisha Vaish. We extend our sincere gratitude to Dr. Peter Blake for serving as a research consultant on this project. We are deeply thankful to the dedicated undergraduate research team for their invaluable assistance with participant recruitment, data collection, prize distribution, and coding: Charlotte Milone, Beverly Lundeen, Rosa Marsharipova, Wenjing Zhang, Katlin Blakeslee, Caleb Soong, Annie Zhu, Madeleine Nguyen, Qianyuan Hu, Georgia Beatty, and Olivia Braye. Finally, we gratefully acknowledge the two children featured in our study stimuli for their tremendous help.

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The analysis codes and materials are available at: https://osf.io/qr2tm/overview?view_only=2dea7c2fce934b088cc954b7c5e84c0b. The data cannot be shared publicly but will be made available upon request to the authors.

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Abstract

Children show early sensitivity to interpersonal inequality, but whether such inequality shapes children's broader social behavior remains unclear. We examined whether experiencing resource disparities – and perceiving those disparities as unfair – impacts children's sharing beyond the immediate inequality context. In Study 1 (N = 240, 60% White), 4- to 9-year-old children and a (pretend) peer played a game in which a machine distributed either half as many tokens to the child (Disadvantageous Inequality, DI), twice as many (Advantageous Inequality, AI), or the same number (Equality, E) as the peer. Children then shared stickers with a new, sick child. Although children at all ages perceived DI as unfair and, with age, also perceived AI as unfair, the inequality conditions did not affect sharing; instead, sympathy towards the recipient predicted children's generosity. In Study 2 (N = 160, 64% White), we manipulated the justifiability of disadvantageous inequality. Six- to 9-year-olds received less than a peer despite winning a game (Unjustified-DI), less after losing (Justified-DI), more after losing (Unjustified-AI), or equal outcomes following a tie (Justified-E). They then shared stickers with a new, neutral child. Children evaluated Justified-DI as fairer than Unjustified-DI and Unjustified-AI, and as fair as Justified-E. Crucially, although the inequality conditions did not significantly impact sharing, children's fairness evaluations predicted their generosity: the more unfair they perceived the distribution, the less they shared. Thus, by school age, children's prosocial behavior may be shaped less by objective resource disparities and more by children's perception of the injustice of those disparities.

Keywords: Inequality, prosocial behavior, fairness perception, development

Public Significance Statement

The present research shows that while young children recognize and feel upset when they experience disadvantageous inequality, this experience alone may not reduce their generosity toward others. Rather, by early school age, children are sensitive to the *fairness* of disadvantageous inequality, and it may be their fairness judgments (alongside social and emotional responses such as sympathy for the recipient) that predict their sharing more than the objective inequality itself. Understanding how perceived fairness shapes children's social behavior within unequal environments may offer insights into reducing the long-term social harms of inequality.

The Impact of Interpersonal Inequality on Children's Prosocial Behavior: The Critical Role of Perceived Unfairness

Children grow up in a world marked by inequality – between families, classrooms, and peers. Research shows that children are remarkably sensitive to such disparities from early in development and have a strong preference for equality (e.g., Fehr et al., 2008; Olson et al., 2012). For example, infants expect to see resources distributed equally and prefer distributors who allocate resources equally (e.g., Buyukozer Dawkins et al., 2019; Geraci & Surian, 2011; Sommerville et al., 2013). Beyond third-party inequality, young children are also sensitive to resource disparities they personally experience. As early as age 3, they show inequality aversion through protest and negative emotional reactions when they receive less than a peer (disadvantageous inequality; Kim et al., 2023; LoBue et al., 2011; Rakoczy et al., 2016). By around age 7, they also view receiving more than others (advantageous inequality) as unfair and object to it (Blake & McAuliffe, 2011; Peretz-Lange et al., 2022; Smith et al., 2013). Children's experiences of inequality further shape whether they subsequently rectify or perpetuate the inequality – with children more likely to rectify with age and when inequalities are perceived as illegitimate (e.g., Chao et al., 2025; Rizzo et al., 2020, 2023).

Despite these early emerging sensitivities and responses, we know little about whether experiencing interpersonal inequality shapes children's broader social behavior. Yet there is reason to think it might. Work with adults suggests that inequality – at both the societal and interpersonal levels – can shape social functioning and behavior. Research on societal inequality has linked perceived inequality to reduced interpersonal trust, heightened competitiveness and self-focus, and lower prosociality (e.g., Côté et al., 2015; Oishi et al.,

2011; Rothstein & Uslaner, 2005; Sommet & Elliot, 2023; Yang & Konrath, 2023). At the interpersonal level, work on inequity aversion and relative deprivation similarly suggests that unfavorable outcomes or social comparisons predict reduced trust, cooperation, and prosocial behavior, and greater risk-taking and competitiveness (e.g., Callan et al., 2017; Gheorghiu et al., 2021; Mishra et al., 2015; Nishi et al., 2015; Zhang et al., 2016). Whether the experience of inequality similarly shapes children's broader social behavior remains an open question.

This is a pressing issue given that wealth and resource inequality have become increasingly pronounced in the U.S. and globally (Pew Research Center, 2020; Chancel, 2022). Children are disproportionately affected by this trend: Some 38% of children (or 27 million children) in the U.S. currently live near or below the poverty line, including nearly 5 million children under the age of 3 (Koball & Jiang, 2018), and worldwide, over 15% of children live in extreme poverty – a rate nearly twice that of adults (Salmeron Gomez et al., 2023). Critically, for many of these children, inequality is likely experienced not as an abstract societal phenomenon but as a concrete, interpersonal reality as they compare their own resources and opportunities to those of peers around them. This more immediate, interpersonal form of inequality was the focus of the present work. Understanding how children respond to such experiences may shed light on how growing inequality shapes children's social development and long-term wellbeing.

The present work asked how interpersonal inequality influences children's prosocial behavior. We focused on prosocial behavior because it is vital for children's social functioning and lifetime wellbeing (Ruck et al., 2019). Prosocial behaviors represent investments in one's social ties that build long-term cooperative relationships and strengthen

social networks, and as such, are integral to children's social bonds and the backbone of healthy families and societies (Laible et al., 2014; Padilla-Walker & Carlo, 2014). Disruptions in children's prosocial capacities can thus prove detrimental to their social development and wellbeing, both concurrently and long-term. Given that experiencing unequal outcomes relative to others has been linked to reduced cooperation and prosocial motivation among adults (e.g., Callan et al., 2017; Nishi et al., 2015; Zhang et al., 2016), prosocial behavior represents a natural starting point for examining whether and how interpersonal inequality may shape children's broader social functioning.

Existing evidence on inequality and children's prosocial behavior

Although little research has directly examined this question among children, existing research has assessed the impact of socioeconomic status (SES) – a broader measure of family income, wealth, and social class – on children's prosocial behavior. Some work suggested that children from lower SES families, compared with children from wealthy families, were more likely to share resources in a lab setting (Chen et al., 2013; Miller et al., 2015). Yet some longitudinal work found the opposite: Children with lower SES backgrounds were more aggressive and less prosocial (Dodge et al., 1994; Malti et al., 2013). A large meta-analysis similarly found that higher social class individuals are more prosocial, and this positive association held consistently for children, adolescents, and adults (Wu et al., 2025). Mixed findings aside, SES reflects broad, cumulative differences in material circumstances that covary with many other factors such as social support or educational background, making it difficult to know whether children perceive and respond to interpersonal disparities per se.

Other work has employed experiments to study children's responses to unequal resource distributions in their social settings. These findings were also mixed. In one study, even 4-year-olds donated less to an unknown, needy child when in a high inequality compared to low inequality environment, even though their own rewards remained constant (Kirkland et al., 2020). However, this effect was not replicated in later studies with 4- to 9-year-olds (Kirkland et al., 2021b; see also Kirkland et al., 2021a). Of note, children in these studies did not report perceiving different levels of fairness across conditions, perhaps indicating that they did not readily notice or interpret ambient inequality in personally meaningful ways. A direct, dyadic experience of inequality relative to a peer may be more salient and more likely to shape children's cognitive and social responses.

Some experimental work has examined whether children's own resource experiences shape their social behavior. For example, children who recalled an experience of resource scarcity shared less generously than children who recalled resource abundance (Ahl et al., 2024). However, this study captured a general sense of having little or plenty (arguably reflecting broader resource availability) rather than dyadic inequality. More recently, Berelejis et al. (2026) found that children who lost a competition, and thus held lower relative status, took more resources from a peer of higher relative status (a previous 'winner' in the same game), whereas children who won did not take more from a peer of lower relative status. These findings suggest that a disadvantaged social position may be particularly salient in shaping children's social behavior. However, this work assessed taking (rather than sharing) resources, which may reflect attempts to rectify inequality rather than prosocial orientations. Moreover, children took resources from peers of known status within the same

immediate context, leaving open whether dyadic inequality shapes children's prosocial behavior more broadly. Addressing this question was the first aim of the present work.

The role of perceived unfairness

Importantly, no prior work has examined the psychological mechanism through which experiences of interpersonal inequality might shape children's prosocial behavior. Specifically, how do children evaluate the fairness of unequal distributions, and how might their fairness perceptions – rather than the objective inequality – shape prosocial behavior?

As reviewed above, children endorse fairness norms and prefer equal distributions from early in development. Critically, however, they do not uniformly object to unequal distributions and even prefer inequality when it is viewed as fair (Starmans et al., 2017). With age, children value equity (distributions based on deservingness such as merit or need) over simple equality (identical distributions) (Rizzo et al., 2016; Schmidt et al., 2016). For example, children allocate more to a recipient who contributed more (Baumard et al., 2012; Schmidt et al., 2016), and by age 7, they favor rectifying resource inequalities by allocating more to the poor (Rizzo & Killen, 2016) and accept unequal distributions if they resulted from a fair procedure (Shaw & Olson, 2014). Thus, children's responses to interpersonal inequality may depend not simply on whether a distribution is unequal, but on whether that inequality is perceived as unfair. This perception may in turn shape their prosocial behavior.

Evidence from adults and adolescents supports this possibility. Among adults, individuals' subjective perceptions of the degrees of inequality are often a stronger predictor of social and psychological outcomes than objective measures of economic inequality (e.g., Du et al., 2024; Gheorghiu et al., 2021; Sommet et al., 2026; see Willis et al., 2022 for

review). Another, more nuanced factor is whether disparities are seen as justified or unjustified. When inequality is perceived to be justified or surmountable, it has weaker negative effects on wellbeing and broader social functioning than when it is perceived to be unjustified or unchangeable (e.g., Alesina et al., 2004; Du & King, 2022; Goudarzi et al., 2020; Schneider, 2012). Perceiving unjustified inequality may have these effects by reducing interpersonal and systemic trust, heightening self-focus, creating a scarcity mindset, or inducing emotions such as moral outrage, resentment, or envy stemming from a sense of personal relative deprivation. (e.g., Callan et al., 2017; Lembregts & Pandelaere, 2014; Oishi et al., 2011). Importantly, all of these responses may extend beyond the individuals in the specific inequality context and dampen social and prosocial motivation towards others more broadly (e.g., Gheorghiu et al., 2021; Greitemeyer & Sagioglou, 2017).

Among adolescents, those who view the world as more just report higher prosocial behavior and lower involvement in bullying (e.g., De Caroli & Sagone, 2014; Wolgast et al., 2023). Further, Chinese adolescents exhibited fewer prosocial behaviors when they perceived higher economic inequality (i.e., their subjective sense of the gap between rich and poor), and critically, system justification mediated this association: Adolescents who viewed the system as unjust were less inclined to act prosocially (Li et al., 2023). It may thus not be objective inequality but rather the perceived fairness or legitimacy of that inequality that shapes adults' and adolescents' social and prosocial behavior.

Although this evidence comes largely from research on societal inequality, which is distinct from dyadic, interpersonal inequality, both forms may nevertheless engage related fairness-based evaluations. In both cases, individuals may evaluate whether the distribution

of resources is just, and experience adverse socioemotional and cognitive outcomes when they perceive it as unjust. Importantly, young children are unlikely to regularly confront or reason about inequality at a societal level. Instead, they encounter inequality most directly in concrete, interpersonal interactions, such as receiving fewer resources than peers. This makes interpersonal inequality a developmentally appropriate and ecologically valid context in which to examine how unequal treatment shapes children's social behavior. The second aim of the present work was thus to examine whether the justifiability of interpersonal inequality, and its perceived fairness, shapes children's prosocial behavior.

Present work

In two pre-registered experiments, we examined how experiencing interpersonal resource disparities affects children's prosocial behavior, and whether the justifiability of those disparities shapes this effect. We relied on a dyadic paradigm in which the participant and a peer received unequal resources, creating an unambiguous sense of relative standing that children are sensitive to from early in development (Blake et al., 2015; Blake & McAuliffe, 2011; LoBue et al., 2011a; Peretz-Lange et al., 2022).

In Study 1, we experimentally created disadvantageous inequality (receiving less than another), advantageous inequality (receiving more than another), or equality of resources among children aged 4 to 9 years, and examined whether this experience impacted children's subsequent prosocial behavior (sharing) towards an unknown peer (adapted from Kirkland et al., 2020). We focused on 4- to 9-year-olds because children's understanding of resource disparities and fairness develops substantially over this age range (e.g., Blake et al., 2015; Elenbaas & Killen, 2016; Yang & Dunham, 2022; Zhang et al., 2021).

In Study 2, with 6- to 9-year-olds, we focused on disadvantageous inequality and manipulated its justifiability. Specifically, we offered children cues to interpret the inequality as justified or unjustified, and examined whether the justifiability shaped children's subsequent prosocial behavior. We limited the age range in Study 2 based on Study 1 results and because the equity understanding that Study 2 relied on becomes more robust during this period (e.g., Rizzo & Killen, 2016). Study 2 focused on disadvantageous inequality because both children and adults in advantaged positions view inequality as fairer than those in disadvantaged positions (Brown-Iannuzzi et al., 2015; Elenbaas, 2019a; Rizzo et al., 2023), and children consistently perceive disadvantageous inequality as unfair across ages (e.g., Peretz-Lange et al., 2022). This makes disadvantageous inequality the most informative context to test whether perceived fairness shapes children's prosocial responses.

In both studies, we measured children's prosociality towards an unknown peer (i.e., unconnected to the inequality experience) to assess whether experiences of interpersonal inequality influence children's *general* prosocial behavior beyond the immediate dyadic context. Generosity towards the same peer or peer of known relative status (as in Berelejis et al., 2026) could index motivation to rectify the imbalance rather than a genuine shift in prosocial motivation. Among adults, and particularly among those in a disadvantaged position, the psychological responses engendered by inequality and relative disadvantage – such as reduced trust, heightened self-focus, a sense of unfairness and resentment – are linked to a widespread reduction in other-oriented motivation, including reduced generosity even towards uninvolved strangers (e.g., Callan et al., 2017; Gheorghiu et al., 2021; Greitemeyer

& Sagioglou, 2017; Oishi et al., 2011). By measuring sharing towards a new, unrelated child, we aimed to test whether inequality exerts a similar generalized effect among children.

We pre-registered two primary hypotheses. First, we expected that children would share more generously with age, as in prior work (Benenson et al., 2007). Second and centrally, we predicted that in Study 1, children who experienced disadvantageous inequality would show less prosocial behavior than children who experienced equality or advantageous inequality, as the negative social outcomes of inequality are often greater among disadvantaged groups (e.g., Gheorghiu et al., 2021; Oishi et al., 2011). However, given the mixed evidence on the effects of relative status (whether measured through SES or dyadic disparities) on prosocial behavior among both children and adults, it was also possible that disadvantageous and advantageous inequality would not differ from each other (e.g., Chen et al., 2013; Malti et al., 2013; Wu et al., 2025). In Study 2, we predicted that children who experienced unjustified disadvantageous inequality may perceive it as particularly unfair and may thus show less prosociality than after experiencing justified disadvantageous inequality.

Additionally, we examined children's evaluations of the fairness of the distributions as a subjective index of how they interpreted the resource inequality, as children may not always interpret the inequality conditions as intended (as in Kirkland et al., 2021b). Although we did not pre-register hypotheses about fairness perceptions, we expected to replicate prior developmental findings that younger children regard disadvantageous inequality as unfair but advantageous inequality as fair, whereas older children also view advantageous inequality as unfair (e.g., Peretz-Lange et al., 2022). We also tentatively expected that perceived fairness

may predict children's sharing, as suggested by prior work with adults and adolescents (Alesina et al., 2004; Li et al., 2023; Schneider, 2012).

Finally, we measured children's emotional responses during the game (both studies) and their sympathy towards the recipient of sharing (Study 1 only). Prior work suggests that even young children show negative emotions when they experience unequal distributions (e.g., LoBue et al., 2011), and that sympathy is a robust predictor of children's prosocial behavior (e.g., Vaish et al., 2009; Williams et al., 2014; Zahn-Waxler et al., 1992). We did not pre-register hypotheses regarding these psychological correlates, but included them to explore whether they related to sharing behavior.

Study 1

Study 1 investigated how the experience of interpersonal inequality impacts 4- to 9-year-old children's general prosocial behavior. To manipulate inequality, children and a (pretend) peer played a distribution game in which a virtual machine appeared to randomly distribute tokens to the participant and the peer (Peretz-Lange et al., 2022). After receiving an equal or unequal distribution, children were given an additional reward of stickers and could share stickers with a new (unfamiliar) child. We also included a non-social disadvantage condition as a control group (see below).

Method

Power analysis. Based on a previous study using a similar paradigm to create equal and unequal distributions across three conditions among 6- to 9-year-old children (Peretz-Lange et al., 2022), a sample size of 112 participants (~ 38 per condition) was sufficient to achieve 80% power to detect small-to-medium effects on children's fairness evaluations (α

= .05) using multiple regression models. However, given that prior work has shown mixed and often small effects of inequality on prosocial behavior among both children and adults (e.g., Kirkland et al., 2020, 2021b; Yang & Konrath, 2023), we anticipated a smaller effect size. Therefore, we conducted an a priori power analysis using G*Power 3.1 (Faul et al., 2007), which indicated that a sample of 54 children per condition (total $N = 216$) would provide 80% power to detect a small effect size ($f^2 = 0.07$) at $\alpha = .05$. To fully counterbalance all aspects of the design (condition, age group, and gender), we increased the sample size to 60 participants in each of four conditions (i.e., 240 total participants) to ensure equal numbers of participants per age and gender group. The procedure and analysis plan for this study were pre-registered on AsPredicted (https://aspredicted.org/5G8_R1N).

Participants. Our final sample included 240 children between 4 and 9 years (range: 4.03-9.92, $M_{age} = 6.98$, $SD = 1.69$, 120 girls, 120 boys, 10 participants per age year per condition)[1]. Participants were recruited from a database of families living in a medium-size mid-Atlantic town as well as from diverse locations in the U.S through Facebook advertisement. Eleven additional participants were excluded: 3 due to experimenter error, 5 because they did not complete the testing session, and 3 because they did not pass comprehension checks (see below).

Geographic information was available for 98% of the sample; this subset came from 35 different U.S. states: 39% were from the Southeast (e.g., VA, GA), 21% were from the Northeast (e.g., NY, PA), 17% from the West (e.g., CA, UT), 17% from the Midwest (e.g.,

[1] Our preregistration incorrectly listed the age range for the study as starting at 5 years due to a typographical error. The correct planned age range was 4 to 9 years.

OH, IL), and 5% from the Southwest (e.g., TX). Demographic information was available for 97% of the sample. Of this subset, 60% identified as White, 15% as Asian, 4% as Hispanic, 3% as Black, and 15% as 'more than one race' or 'other'. Finally, 97% of families reported income and education. The median household income level was $100,001-$150,000, and 99% of participants in the subset had at least one parent with a bachelor's degree or higher. The research was approved by the University of Virginia Institutional Review Board.

Materials and procedure. Participants were tested online via Zoom due to the COVID-19 pandemic. Participants' parents were asked to help participants set up the Zoom meeting and let their children answer all questions by themselves. The experimenter (E) then briefly explained the study to children to make sure they felt comfortable and provided assent before starting the study. Next, E shared their screen to present study slides to participants. The testing sessions were recorded using the 'record' function on Zoom.

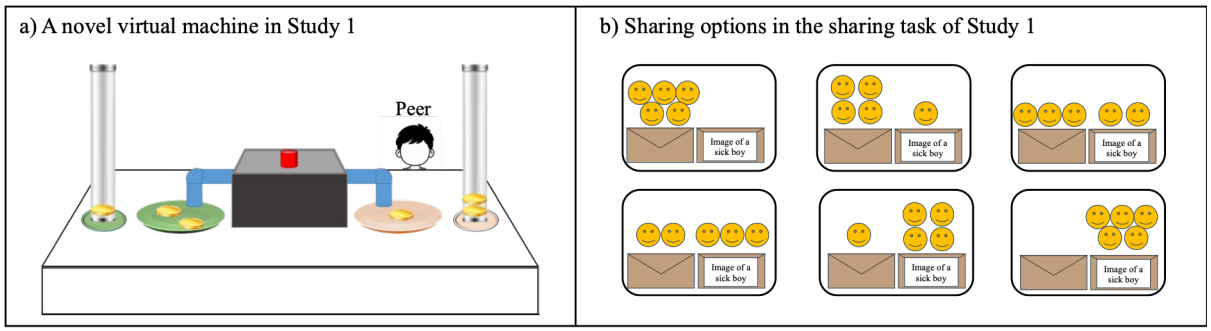
The study consisted of two main phases: manipulation of interpersonal inequality followed by the prosocial task.

Manipulation of interpersonal inequality. In the first phase, children played a resource distribution game online with a gender-matched (pretend) peer. To increase believability, a prerecorded video of a real child was presented in which the child introduced themselves by name and said they were ready to play the game. Participants were told a machine would distribute tokens between them and the peer (participants did not complete any tasks to earn these tokens). They were then presented with a novel virtual machine that appeared to randomly give out tokens to participants and the peer (see Figure 1a). To make the resources valuable for participants, they were told that at the end of the game, they could

exchange the tokens for real gifts (i.e., scratch art papers, which can be scratched with a bamboo stylus to create colorful artwork) that E would mail to them after the game.

Figure 1

Novel Virtual Machine and Sharing Task in Study 1



Note. (a) Example scenario (boy version) showing the novel virtual machine, which distributed tokens to the participant (left) and a pretend peer (right). For each round, the tokens appeared on the plates, then moved to the corresponding tubes. One full tube (12 tokens) could be exchanged for one scratch paper. (b) Example scenario (boy version) showing all 6 options for sharing stickers with a sick peer. The specific number of stickers was displayed above the participant's envelope (left) and the peer's envelope (right). The original sick-peer image is omitted here for copyright reasons.

Over 10 rounds, the machine was pre-set to sometimes dispense equal amounts to the participant and peer, sometimes more to the participant, and sometimes less to the participant. At the end of the 10 rounds, the participant either had the same number of tokens as the peer (Equality condition, E), twice that of the peer (Advantageous Inequality condition, AI), or half that of the peer (Disadvantageous Inequality condition, DI). We controlled the total number of tokens and rewards participants received across conditions (i.e., 24 tokens for 2 scratch papers) and only changed the number of tokens and rewards the peer received to achieve these allocations. To ensure that any effects of resource inequality arose from social comparisons instead of mere frustration about receiving less, we also conducted a Nonsocial-Disadvantage (NS-D) condition in which the machine distributed tokens between the participant and a box. Participants were told that the tokens distributed to the box would be

put into the box, and no one would get those tokens. Children were randomly assigned to one of the four conditions (E, AI, DI, or NS-D). To ensure comprehension, on each of the 10 distribution rounds, children were asked how many tokens they received and whether they received more or less than the peer. If they answered incorrectly, E helped participants count the tokens and made sure they knew the correct number of tokens and understood who received more or less. After re-counting, if they still answered incorrectly, the experimenter provided them with the correct information. Children who answered incorrectly after re-counting on 3 or more rounds (n = 3) were excluded from analyses.

Sharing task. In the second phase we measured children's sharing towards a new child. E first told children that since they did so well, in addition to the scratch papers, she would also mail 5 stickers to them. To maximize children's interest in the stickers, we asked participants to pick their favorite from three types of stickers (stars, happy faces, and hearts). Participants were then presented with a picture of a sick child (White, gender-matched) and told that if they wanted, they could share some of their stickers with the sick child to make him/her happy. Previous work indicates that by the age of 5, children's prosocial behavior is impacted by reputational concerns (e.g., Engelmann et al., 2012). To reduce reputational concerns, E told children that she would leave the room while they made their sharing decision and only the computer would know their decision. E then stepped out of participants' view and played a pre-recorded audio with a computerized female voice to make it appear as though the computer alone was conducting the sharing task.

On the screen, children were presented with 6 choices of sharing options ranging from sharing none of the stickers to sharing all 5 stickers (see Figure 1b), and the computerized

voice asked children to say their decision out loud. After children said their decision, E (off screen) selected the slide corresponding to children's sharing decision. The computerized voice then asked children to confirm that the number of stickers they wanted to share was correct. If children did not confirm, E (still off screen) returned to the slide that presented the 6 sharing options and repeated the steps. After children confirmed their sharing decision, they were shown the rest of the stickers going into their own envelope and both envelopes closing. Then E returned to the computer and continued the remaining session. Thus, from the participants' perspective, E did not know what the child had selected. We used the number of stickers that participants shared as the main dependent variable, ranging from 0 to 5.

Social-emotional measures. In addition to the main measure of sharing, we measured three social-emotional responses that may be influenced by interpersonal inequality.

Emotional changes. We measured children's emotional reactions before and after they played the distribution game. Each time, children were presented with a 5-point scale of faces (from very sad to very happy), and asked to point to the face that showed their current feeling. Children's responses were coded from 1 ("very sad") to 5 ("very happy"). For each participant, we computed participants' emotional change (post-test emotion score minus pre-test emotion score), with values ranging from -4 to 4. Negative emotional change scores indicated children were sadder after playing the distribution game, positive scores indicated children were happier after playing the game, and 0 indicated no change.

Sympathy. After the sharing task, we measured children's sympathy towards the sick child by asking how sorry they feel for the child on a 3-point scale (1 = "not at all sorry" to 3

= "really sorry"). This measure was included to assess sympathy, which is an important motivator of children's sharing behavior (e.g., Vaish et al., 2009).

Fairness evaluation. After the sympathy measurement, we asked children to evaluate the fairness of the virtual machine in two steps. We first asked a forced-choice question ("*Do you think the machine is fair or not fair?*"). Depending on children's response to this question, we asked an additional question to measure how fair/unfair on a 3-point scale, from "a little bit (not) fair" to "really (not) fair". We combined these two fairness evaluation questions to produce a score from 1 to 6 (1 = "really not fair", 6 = "really fair").

Debriefing. After the entire testing session, E told children in the DI condition that the machine had malfunctioned and that she would give them 2 more scratch papers; thus, children in the DI condition ultimately received the same number of scratch papers as the peer. After the study, the scratch papers and stickers were mailed to all participants.

Transparency and Openness. Analysis codes and research materials for both studies are available at https://osf.io/qr2tm/overview?view_only=2dea7c2fce934b088cc954b7c5e84c0b (Shu & Vaish, 2026). Due to ethical considerations and to protect the privacy of participating children and families, the data cannot be shared publicly but will be made available upon request to the authors.

Results

Analytic strategy. We performed a linear regression model on the number of stickers participants shared as a function of condition (AI, DI, E, or NS-D), age (continuous and

centered), and their interactions in R (Version 4.4.3, R Core Team, 2025).[2] The interaction term was dropped from the final model if it was not significant. For the three social-emotional measures (fairness, emotional changes, and sympathy), we conducted similar linear regression models. We then conducted exploratory analyses, as outlined in our preregistration, to examine the relations between children's sharing behavior and the social-emotional measures. In addition, based on prior work among adolescents (Li et al., 2023), we explored whether children's perceived fairness mediated the influence of condition on sharing. These exploratory models focused on the three interpersonal conditions (AI, DI, and E) to specifically examine the effects of interpersonal inequality. Results from equivalent analyses including all four conditions are reported in the Supplementary Materials. We also conducted pre-registered exploratory analyses by submitting participant gender and SES as covariates in the general model on children's sharing behavior. These exploratory analyses revealed a similar pattern of results as the main analyses (see Supplementary Materials).

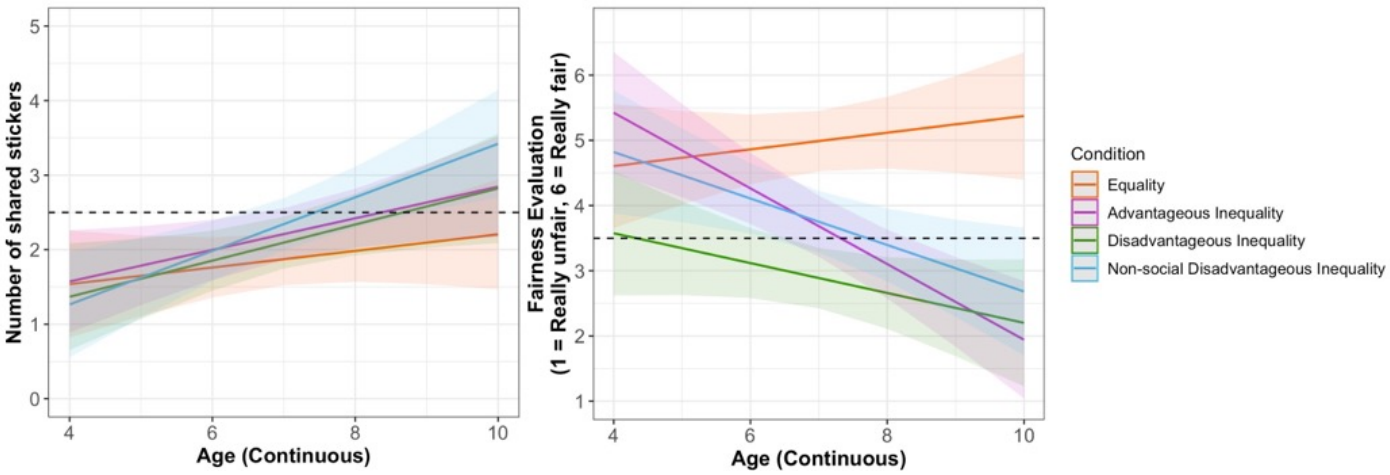
Sharing behavior. The linear regression model did not reveal any significant interaction contrasts (all *p*s > .10), thus the interaction term was dropped. The final model showed a significant main effect of age ($B = 0.23$, $SE = 0.05$, 95% CI [0.13, 0.33], $p < .001$), suggesting that with age, children were more likely to share stickers with the sick child (Figure 2, left). This is in line with previous findings that older children are more generous and willing to share their own resources with others (e.g., Benenson et al., 2007; Kirkland et

[2] Note that in our pre-registration, we inadvertently proposed to run a multilevel mixed-effects linear regression model on children's sharing behavior as a function of condition, age, and their interactions, with a random intercept for participants. In fact, our design did not fulfill the requirement to run a multilevel mixed-effects linear regression model; we therefore conducted a simple linear regression model.

al., 2021b). However, contrary to our predictions, children who received less than the peer (DI condition: $M = 2.08$, $SD = 1.27$) did not share significantly less from children who received more than the peer (AI condition: $M = 2.22$, $SD = 1.38$) or in Equality condition ($M = 1.87$, $SD = 1.49$) (both $ps > .38$). Children's sharing in E and AI conditions also did not differ from one another ($p = .19$). Thus overall, we did not find evidence that children's sharing behavior towards a new, unknown peer was impacted by their experience of interpersonal inequality. Children's sharing in the NS-D condition also did not differ significantly from the other conditions (all $ps > .06$).

Figure 2

Children's Prosociality and Fairness Evaluations by Age and Condition in Study 1



Note. Left: The number of stickers that children shared across age and condition in Study 1; Right: Children's fairness evaluation of the machine across age and condition in Study 1. The lines show the predicted values from linear regression models. Shaded areas represent 95% confidence intervals.

Fairness evaluations. A linear regression model on children's fairness evaluations as a function of condition, age, and their interaction revealed a significant main effect of condition (Figure 2, right). As expected, children in E condition ($M = 4.98$, $SD = 1.71$) regarded the machine as quite fair. However, children in the three inequality conditions

viewed the machine as significantly less fair than in E (DI condition: $M = 2.90$, $SD = 1.94$; $B = -2.09$, $SE = 0.33$, 95% CI [-2.74, -1.44], $p < .001$; AI condition: $M = 3.64$, $SD = 2.00$; $B = -1.29$, $SE = 0.33$, 95% CI [-1.95, -0.63], $p < .001$; NS-D condition: $M = 3.78$ $SD = 2.01$; $B = -1.23$, $SE = 0.34$, 95% CI [-1.88, -0.57], $p < .001$).

The model also revealed significant condition × age interaction contrasts (AI: $p < .001$; NS_D: $p = .017$). To interpret this interaction, we conducted simple slopes analyses using estimated marginal trends for each condition with the *emtrends* function in R. Results showed that with age, children were more likely to judge the machine as unfair in the AI ($B = -0.58$, SE = 0.13, 95% CI [-0.84, -0.32], $p < .001$) as well as NS_D conditions ($B = -0.36$, SE = 0.14, 95% CI [-0.64, -0.08], $p = .013$). In contrast, age did not significantly predict fairness evaluations in the DI ($B = -0.23$, SE = 0.14, 95% CI [-0.51, 0.05], $p = .11$) or E conditions ($B = 0.13$, SE = 0.14, 95% CI [-0.15, 0.41], $p = .37$).

Emotional changes. We submitted children's emotional change scores to a linear regression model as a function of condition, age, and their interaction. We found no significant interaction contrasts in this model (all ps > .44), so the interaction term was dropped. The final model revealed no significant effect of age ($p = .15$) but a main effect of condition. As expected, children showed negligible emotional change in the E condition ($M = 0.10$, $SD = 0.86$). Similarly, children in AI condition also did not show much emotional change ($M = 0.07$, $SD = 0.97$), and this did not differ from E condition ($p = .83$) (see Figure 1S, left, in the Supplementary Materials). In contrast, children who experienced disadvantage, whether social or non-social, showed significantly more negative emotional change than in E condition: DI ($M = -0.59$, $SD = 1.10$; $B = -.69$, $SE = 0.18$, 95% CI [-1.04, -0.35], $p < .001$)

and NS-D (M = -0.37, SD = 0.90; B = -0.47, SE = 0.18, 95% CI [-0.81, -0.12], p = .009). The DI and NS-D conditions did not differ significantly from one another (p = .20).

To further examine whether the emotional changes in DI and NS-D indicated that children in these conditions indeed became significantly less happy after the game, we ran one sample t-tests to compare the emotional changes against the neutral level (i.e., 0, or no emotional change). In line with our hypotheses, experiencing disadvantage negatively influenced children's emotions such that children reported being significantly less happy after they received less than the peer (DI: $p < .001$) or the box (NS-D: p = .003).

Sympathy. A similar linear regression model on children's sympathy towards the new (sick) child revealed no significant interaction contrasts (all ps > .56). After dropping the interaction term, we found no significant main effect of participant age (p = .26) or condition (ps > .25). Instead, children across the age range and the four conditions generally reported feeling '*really sorry*' for the sick child on the 1-3 Likert scale (AI: M = 2.65, SD = 0.58; DI: M = 2.52, SD = 0.60; E: M = 2.52, SD = 0.65; NS-D: M = 2.53, SD = 0.65), suggesting that participants felt strong sympathy towards the sick child regardless of age or condition.

Exploratory analyses. Our primary analyses above included all four conditions. However, we conducted additional exploratory analyses focusing only on the three *interpersonal* conditions (AI, DI, and E), excluding the non-social control condition (NS-D). We reasoned that including NS-D, which lacked peer interaction or comparison, into our models might obscure potential effects of interpersonal inequality on children's social-emotional responses. By focusing on the three interpersonal conditions, we aimed to more directly examine how interpersonal inequality involving social comparison influences

children's judgments and behavior. We explored: 1) whether children's sharing behavior was related to the three social-emotional measures; and 2) if fairness perceptions were associated with sharing, then whether they mediated the influence of condition on sharing.

Pearson correlations between children's sharing and each social-emotional measure revealed that sharing was positively correlated with sympathy (r = .21, 95% CI [.06, .35], p = .006), but not with fairness evaluations (r = - .01, 95% CI [- .16, .14], p = .90) or emotional changes (r = - .10, 95% CI [- .24, .05], p = .21). Since fairness evaluations did not predict sharing, we did not conduct a further mediation analysis as described above.

Discussion

Study 1 examined how interpersonal inequality influences 4- to 9-year-old children's general prosocial behavior. We found, first, that our manipulation of resource inequality successfully shaped children's fairness perceptions and replicated developmental changes seen in previous work (Blake & McAuliffe, 2011; Peretz-Lange et al., 2022). Specifically, both younger and older children viewed DI as unfair, but only older children also perceived AI as unfair. Additionally, our inequality manipulation impacted children's emotional responses such that children reported feeling less happy after receiving less than the peer (or box in NS-D) but not after receiving the same or more than the peer. Nonetheless, contrary to our predictions, experiencing disadvantageous inequality did not undermine children's prosociality. Instead, sharing increased with age, and was predicted by sympathy towards the sick recipient, consistent with a substantial body of evidence that sympathy is a strong and robust motivator of children's prosociality (e.g., Malti et al., 2016; Vaish et al., 2009).

These findings may indicate that dyadic resource inequality does not shape children's broader prosocial behavior. Alternatively, perhaps children's prosocial behavior is impacted by resource inequality but in Study 1, sympathy towards the sick recipient may have been a strong enough motivator of prosocial behavior that it effectively overrode any effects that resource inequality might otherwise have had on children's prosocial behavior. Because sympathy for the recipient was not the primary focus of our investigation, Study 2 replaced the sick recipient with a neutral one. It is also possible that the disadvantageous inequality in Study 1 was not perceived as sufficiently unjust to alter children's prosocial behavior, as described below. Study 2 was designed to address this possibility.

Interestingly, while children across ages found both NS-D and DI conditions to be emotionally negative, only older children also evaluated NS-D condition as unfair. Children's perceptions of inequality thus changed with age: Younger children reacted emotionally to unequal outcomes, but with age, children grasped that the system of distribution (the machine) was unfair no matter whether the recipient was a human or nonsocial entity. Children thus gradually become attuned to the fairness of the distribution process rather than simply the outcomes of distributions (see Dunham et al., 2018).

Study 2

In Study 1, 4- to 9-year-old children recognized disadvantageous inequality and felt upset at being disadvantaged, but experiencing this inequality did not undermine their prosociality towards general others. Instead, only children's sympathy towards the sick recipient predicted their prosocial behavior, perhaps because it overrode any effects of disadvantageous inequality. An important further possibility is that the disadvantageous

inequality in Study 1 was not interpreted as sufficiently unjust to alter children's behavior. Because children received resources without any performance context or explanation for the unequal outcome, they may have attributed the disadvantage to bad luck, randomness, or the mechanics of the game and therefore justified their disadvantage. As a result, although they perceived the disadvantage, the experience may not have elicited a strong enough sense of injustice to influence their subsequent prosocial behavior, as suggested by the non-significant correlation between fairness evaluations and prosocial behavior. Indeed, in prior work with adults and adolescents, it was the perceived fairness or justifiability of inequality, rather than the inequality itself, that more strongly predicted emotional and behavioral responses such as prosociality (e.g., Alesina et al., 2004; Li et al., 2023; Schneider, 2012).

To address these concerns, Study 2 introduced some important changes. First, we manipulated the justifiability of disadvantageous inequality – defined as whether the outcome was deserved based on task performance – to more strongly shape children's perceptions of fairness. Children either received fewer resources despite outperforming their peer (Unjustified-DI) or due to poorer performance (Justified-DI). This allowed us to assess the potential causal role of unfair disadvantage in shaping children's prosocial responses. We also included Unjustified-AI as well as E conditions for comparison.

Second, to more clearly isolate the effects of interpersonal inequality on children's prosocial behavior, we replaced the sick recipient in the sharing task with a neutral recipient to dampen the strong sympathy that may have overshadowed the effects of inequality in Study 1. Finally, we modified the age range to 6–9 years. This was because Study 2 focused on the justifiability of inequality, which required an understanding of equity and procedural

fairness, and prior work as well as Study 1 findings suggest that younger children are less sensitive to these aspects and more focused on outcomes.

We hypothesized (pre-registered) that children experiencing unjustified inequality would perceive it as more unfair compared to justified inequality or equality, which would lead them to show reduced prosociality. We tentatively expected (not pre-registered) that this effect would be more apparent in the Unjustified-DI condition than Unjustified-AI condition, as only unjustified disadvantage is likely to evoke adverse socioemotional and cognitive outcomes (e.g., Alesina et al., 2004; Du & King, 2022). Based on the results of Study 1, we also expected (pre-registered) that prosocial behavior would increase with age.

Method

Power analysis. Study 2 employed four conditions (two justified and two unjustified conditions). Based on previous work (Peretz-Lange et al., 2022), a sample size of 38 participants per condition would be sufficient. Additionally, a post-hoc power analysis of the linear regression on children's prosocial behaviors in Study 1 as the function of 4 conditions, age, and their interaction indicated that to detect the same effect size of 0.10 with 80% power and an alpha level of 0.05, a sample of 135 participants was required (G*Power 3.1; Faul et al., 2007). To fully counterbalance all aspects of the design (i.e., condition, participant age, and gender), we pre-registered a sample size of 160 (40 children per condition). Procedures and analysis plans for this study were pre-registered (https://aspredicted.org/5gmy-5qd5.pdf).

Participants. Our final sample size included 160 children between 6 and 9 years (range: 6.01-9.91, M_{age} = 8.00, *SD* = 1.09, 80 girls, 80 boys, 10 participants in each age year and condition). Participants were recruited from a database of families living in a medium-

size mid-Atlantic town as well as from diverse locations in the U.S. through Facebook advertisement. Three additional participants were excluded: 1 due to experimenter error and 2 because they were distracted and did not complete the study.

Geographic information was available for the full sample. Participants resided in 34 different U.S. states: 51% were from the Southeast (e.g., VA, TN), 21% were from the Northeast (e.g., NY, MA), 11% from the West (e.g., CA, UT), 10% from the Midwest (e.g., IL, MN), and 7% from the Southwest (e.g., AZ). Demographic information was available for 96% of the sample. Of this subset, 64% identified as White, 16% as Asian, 8% as Hispanic, 1% as Black, and 12% as 'more than one race' or 'other.' Finally, 96% of families reported income and education. The median household income level was $100,001-$150,000, and 95% of the participants had at least one parent with a bachelor's degree or higher.

Materials and procedure. To maintain consistency in the testing environment across both studies, Study 2 was also conducted online via Zoom. Study 2 adapted methods of Study 1 with the following differences. First and most importantly, the disadvantageous inequality that children experienced was either justified or unjustified. Participants were presented with the same virtual machine as in Study 1. However, the machine was described as a rewarding machine that would give more rewards to the winner of a challenging game (finding specific objects hidden in a picture). Children then played the game against a pretend (gender-matched) peer. Regardless of children's real performance, in the justified disadvantage condition, children were told they lost the game and received half as many rewards as the peer (Justified-DI). In the unjustified disadvantage condition, children were told they won the game but still received half as many rewards as the peer (Unjustified-DI). To assess whether

any effects we found were specific to DI or applied more generally to inequality, we also conducted an unjustified advantage condition in which children lost the game but then received twice as many rewards as the peer (Unjustified-AI). Finally, we also conducted a justified equality condition as the baseline comparison, in which participants were told that they tied with their peer and received equal rewards (Justified-E). Participants were randomly assigned to one of these four conditions. Each participant played two rounds of a hidden object game, and on each round, they experienced the same condition.

Second, children were introduced to a new (gender-matched) child and told that the child could not play this game but really liked stickers. Children were asked whether they would like to share their stickers with this child. If they said 'yes,' the identical sharing task was presented as in Study 1. If they said 'no,' they were not presented with the sharing task and were given a sharing score of 0. Children's emotions before and after the distribution game and their fairness evaluations of the rewarding machine were measured as in Study 1.

Results

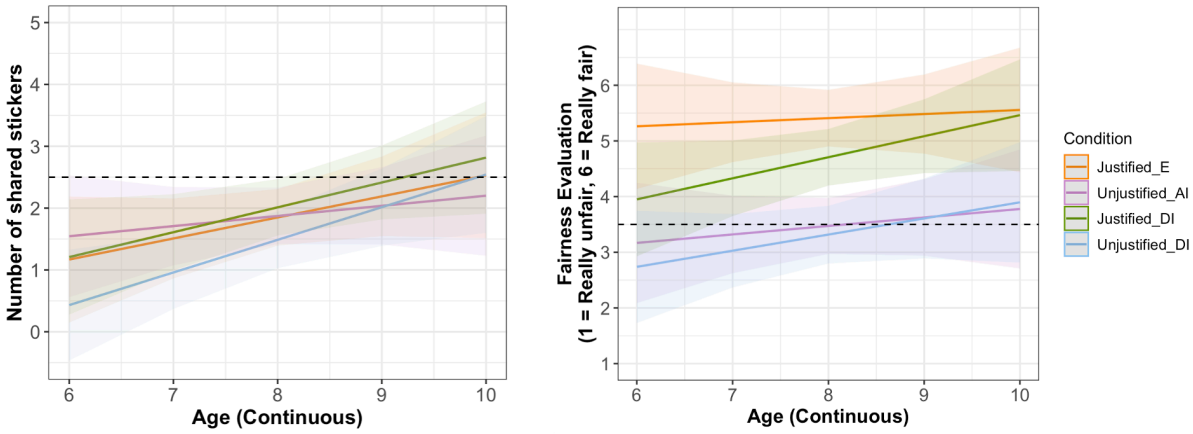
Analytic strategy. The analytic approach was identical to Study 1. We performed linear regression models separately on children's sharing behavior, emotion change, and fairness evaluation as a function of condition (Justified-DI, Unjustified-DI, Unjustified-AI or Justified-E), age (continuous and centered), and their interactions in R (Version 4.4.3, R Core Team, 2025). We then conducted identical exploratory analyses as in Study 1 (see Supplementary Materials for models including participant gender and SES).

Sharing behavior. The linear regression model again did not reveal significant interaction contrasts (all ps > .54), thus the interaction term was dropped. Like in Study 1,

with age, children shared more stickers with the new peer (B = 0.37, SE = 0.11, 95% CI [0.16, 0.57], p < .001). Comparing across conditions, we found that although, as expected, children who won the game but received fewer rewards (Unjustified-DI) shared the fewest stickers (M = 1.45, SD = 1.50), this was not significantly different from the Justified-E condition (M = 1.85, SD = 1.5) or Unjustified-AI condition (M = 1.88, SD = 1.38) (both ps > .22) (Figure 3, left). Children's sharing in the Unjustified-DI was somewhat lower than in the Justified-DI condition (M = 2.03, SD = 1.61), but this contrast also did not reach significance (B = -0.54, SE = 0.32, 95% CI [-1.17, 0.10], p = .10). No other condition contrasts were significant (all ps > .61). Thus, overall, though the means were in the predicted directions, children's sharing towards a new, unknown peer was not significantly undermined by their experience of disadvantageous inequality even when this inequality was unjustified.

Figure 3

Children's Prosociality and Fairness Evaluations by Age and Condition in Study 2



Note. Left: The number of stickers that children shared across age and condition in Study 2. Right: Children's fairness evaluations of the machine across age and condition in Study 2. The lines show the predicted values from linear regression models. Shaded areas represent 95% confidence intervals.

Fairness evaluations. A similar linear regression model on children's fairness evaluations did not reveal significant interaction contrasts (all ps > .36). After dropping the

interaction term, the model indicated that with age, children viewed the rewarding machine as fairer (B = 0.24, SE = 0.12, 95% CI [0.01, 0.46], p = .046). Across the age range, compared with children in Justified-E (M = 5.41, SD = 0.97), children in Unjustified-DI (M = 3.29, SD = 1.89) and Unjustified-AI (M = 3.48, SD = 1.77) viewed the machine as significantly less fair (Unjustified-DI: B = -2.10, SE = 0.36, 95% CI [-2.81, -1.38], p < .001; Unjustified-AI: B = -1.94, SE = 0.36, 95% CI [-2.64, -1.23], p < .001). Critically, however, children in Justified-DI (M = 4.72, SD = 1.65) regarded the machine as significantly fairer than children in the two Unjustified conditions (Unjustified-DI: B = -1.40, SE = 0.36, 95% CI [-2.11, -0.69], p < .001; Unjustified-AI: B = -1.24, SE = 0.36, 95% CI [-1.94, -0.54], p < .001) and viewed it to be as fair as children in Justified-E (B = -0.70, SE = 0.36, 95% CI [-1.40, 0.01], p = .06) (Figure 3, right). Thus, in line with our hypotheses, children across this age range did not simply view all disadvantage as equally unfair; rather, they differentiated the justified from the unjustified disadvantage and regarded justified DI as fairer than unjustified DI.

Emotional changes. We submitted children's emotional change scores into a linear regression model as a function of condition, age, and their interactions. Since there were no significant interaction contrasts (all ps > .41), the interaction term was dropped. The final model indicated a main effect of age such that with age, compared with their pre-game emotions, children showed more positive emotions after they played the game (B = 0.15, SE = 0.07, 95% CI [0.01, 0.28], p = .036). With respect to the conditions, similar to Study 1, children showed negligible emotional change in the Justified-E condition (M = 0.05, SD = 0.71) and Unjustified-AI condition (M = 0.23, SD = 1.05), with no significant difference between the two (p = .42). As expected, children in both DI conditions showed reduced

happiness (Justified-DI: M = -0.53, SD = 1.08; Unjustified-DI: M = -0.26, SD = 1.00) and did not significantly differ from one other (p = .21). When compared to Justified-E, unexpectedly, only the emotional change in Justified-DI was significantly lower (B = -0.58, SE = 0.21, 95% CI [-1.00, -0.16], p = .007). The emotional change in Unjustified-DI did not differ from Justified-E (B = -0.31, SE = 0.22, 95% CI [-0.73, 0.12], p = .16) but only differed from Unjustified-AI (B = 0.48, SE = 0.22, 95% CI [0.06, 0.90], p = .027) (Figure 1S, right). One sample t-tests to assess whether the emotional changes significantly differed from 0 (i.e., no emotional change) indicated that only children in Justified-DI condition were significantly less happy after the game (p < .003); children in Unjustified-DI condition did not report a significant decrease in happiness (p = .12).

Thus, children generally reported more negative emotions after disadvantageous inequality, whether justified or unjustified. However, emotions were more substantially negative in the Justified-DI condition. This may be because children in Justified-DI experienced two upsetting events: losing the games and also receiving fewer rewards, whereas children in Unjustified-DI had the positive experience of winning the games, which may have reduced some of the negative emotion resulting from the unjustified distribution.

Exploratory analyses. As pre-registered, we examined: (1) whether fairness perceptions and emotional changes predicted children's sharing; and (2) whether condition effects on sharing may be mediated by fairness perceptions. These analyses focused on three conditions: Justified-DI, Unjustified-DI, and Justified-E. The Unjustified-AI condition was excluded because it involved a qualitatively different experience of inequality (i.e., receiving more), which is less likely to elicit a sense of injustice alongside other socioemotional and

cognitive outcomes (e.g., Alesina et al., 2004; Du & King, 2022), and thus did not inform our hypothesis about how the experience of *disadvantage* and its justifiability might influence children's fairness judgments and subsequent prosocial behavior (see Supplementary Materials for analyses with all four conditions).

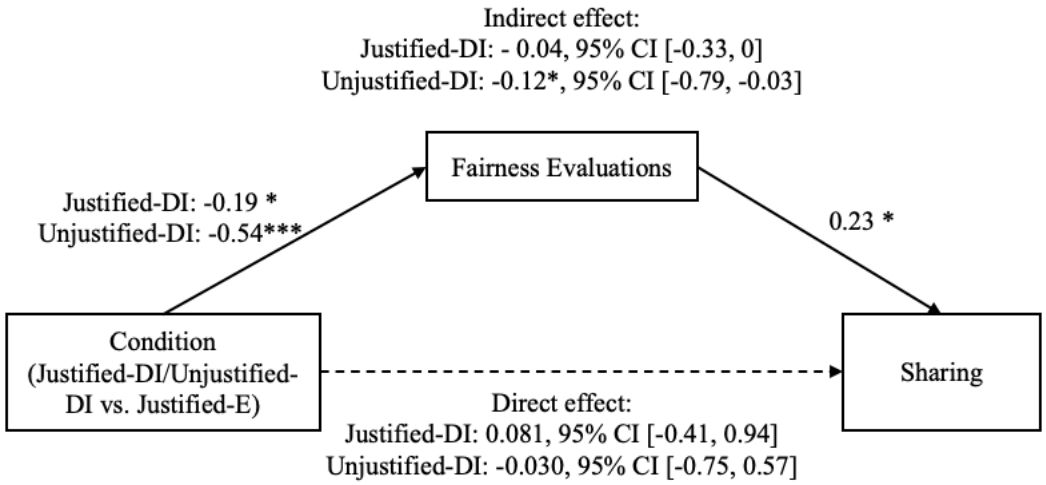
Pearson's correlations revealed that children's fairness perceptions were positively correlated with their sharing behavior, $r = .29$, 95% CI [.11, .45], $p = .002$. That is, the more fair or justifiable children perceived the distributions to be, the more generously they shared. As in Study 1, children's emotional changes were not significantly correlated with their sharing behavior, $r = .15$, 95% CI [- .03, .33], $p = .10$.

We then conducted a mediation analysis to test whether children's fairness perceptions explained the association between condition and sharing behavior. We conducted the mediation of fairness evaluations (mean-centered) using the *sem* function in the *lavaan* package (Rosseel, 2012) in R. Indirect effects were estimated using nonparametric bootstrapping with 10,000 resamples and bias-corrected 95% confidence intervals. The model included three conditions (reference group: Justified-E) and participant age (mean-centered) as a covariate. Partially in line with our prediction, the indirect effect of the Unjustified-DI condition on sharing via fairness perception was significant ($B = -0.12$, $SE = 0.19$, $p = .042$, 95% CI [-0.79, -0.04]), suggesting that children who experienced unjustified disadvantage perceived it as less fair, which in turn reduced their prosocial behavior (Figure 4). The indirect effect of the Justified-DI condition was not significant ($B = -0.04$, $SE = 0.09$, $p = .12$, 95% CI [-0.33, 0.00]). Notably, neither condition had a significant direct or total effect on

sharing behavior (*p*s > .12), indicating that fairness perception may play a critical role in explaining the reduction in sharing in the Unjustified-DI condition.

Figure 4

Mediation of the Effect of Condition on Sharing via Fairness Evaluations in Study 2



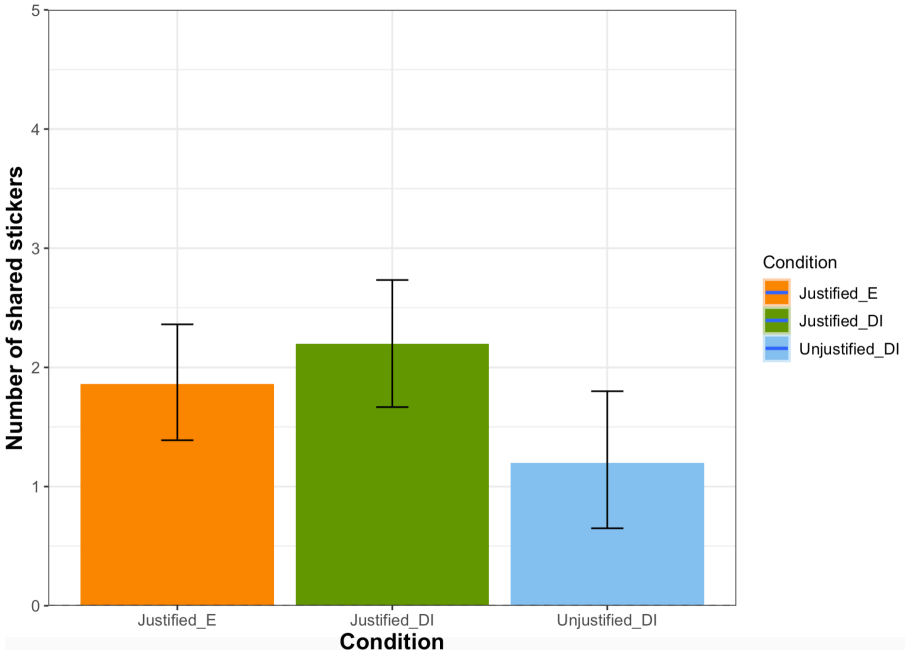
Note. Solid arrows represent significant paths ($p < .05$); dashed arrows represent nonsignificant direct effects. Path coefficients are standardized estimates. Indirect effects represent the influence of condition on sharing behavior via children's fairness evaluations. Age was included as a covariate predicting both fairness evaluations and sharing behavior but is not depicted in the figure for clarity.
$p < .05$ *, $p < .01$ **, $p < .001$ ***.

Exploratory subgroup analysis. Given that the exploratory analyses pointed to the potentially key role of fairness perceptions on children's sharing, we further probed the role of this factor with an exploratory regression analysis on the subset of participants in the DI and E conditions whose fairness judgments aligned with the intended manipulation: children in Justified-DI condition who evaluated the outcome as fair ($n = 30$), Unjustified-DI as unfair ($n = 20$), and Justified-E as fair ($n = 36$). This approach allowed us to more directly assess whether perceived fairness, rather than condition assignment alone, influenced sharing (See Supplementary Materials for similar findings with all four conditions).

A linear regression predicting children's sharing as a function of condition, age, and their interaction did not reveal significant interaction contrasts (*p*s > .75). After dropping the interaction term, age was marginally significant ($B = 0.29$, $SE = 0.15$, $p = .053$, 95% CI [0.00, 0.57]). Critically, the final model revealed a significant condition difference: Children in Unjustified-DI who viewed the outcome as unfair ($M = 1.20$, $SE = 0.30$) shared significantly less than those in Justified-DI who viewed it as fair ($M = 2.20$, $SE = 0.28$; $B = 0.89$, $SE = 0.43$, $p = .041$, 95% CI [0.05, 1.74]). Sharing in Justified-E ($M = 1.86$, $SE = 0.26$) did not differ from Justified-DI ($B = 0.28$, $SE = 0.37$, $p = .455$, 95% CI [-0.44, 1.00]) (Figure 5). These exploratory results offer converging evidence that subjective fairness perceptions may be a key mechanism underlying children's prosocial responses under inequality.

Figure 5

Children's Sharing by Condition among Subgroup in Study 2



Note. Figure represents subset of participants whose fairness judgments aligned with their respective conditions (Justified-DI: perceived as fair; Unjustified-DI: perceived as unfair; Justified-E: perceived as fair). Error bars represent 95% confidence intervals.

Discussion

Study 2 examined whether the perceived justifiability of disadvantageous inequality impacts 6-9-year-old children's prosocial behavior. We found that children across the age range did incorporate justifiability into their fairness judgments: They viewed justified disadvantage to be as fair as equality, whereas they perceived both unjustified disadvantage and unjustified advantage to be unfair. However, the experimental manipulation of justifiability itself did not produce significant differences in children's sharing behavior.

Crucially, exploratory analyses revealed that this null effect may be due to individual variation in how children subjectively interpreted the fairness of inequality. Specifically, the mediation analysis revealed that the more unfair children perceived the disadvantage to be, the less generously they shared. Moreover, when we focused our analysis on participants whose fairness judgments aligned with the intended manipulation (i.e., who perceived Justified-DI and Justified-E as fair, and Unjustified-DI as unfair), condition differences in prosocial behavior did emerge: As predicted, children in the Unjustified-DI condition shared less than those in the Justified-DI condition, despite both groups receiving objectively fewer resources. Together, these findings emphasize the potentially critical role of children's subjective interpretations of inequality in their subsequent generosity.

General Discussion

In two experimental studies, we examined how experiencing interpersonal resource inequality – and its justifiability – shapes broader prosocial behavior in early and middle childhood. In Study 1, 4- to 9-year-old children viewed disadvantageous inequality as unfair (replicating prior work; Blake et al., 2015; Peretz-Lange et al., 2022), and reported negative affect after being disadvantaged, but this experience did not undermine their prosocial

behavior. Instead, prosociality was predicted only by sympathy for the recipient (who was described as being sick). In Study 2, after replacing the sympathy-inducing recipient and manipulating the justifiability of disadvantage, we found that 6- to 9-year-old children's fairness judgments varied based on whether the disadvantage was deserved: They viewed justified disadvantage (receiving less due to poorer performance) as fair but unjustified disadvantage (receiving less despite better performance) as unfair. Crucially, these subjective fairness evaluations, rather than the objective disadvantage, predicted children's subsequent sharing: the less fair they perceived the inequality, the less generously they shared with a new peer. Together, these findings suggest that subjective perceptions of fairness may be more proximal predictors of children's prosocial behavior than objective resource disparities alone.

This pattern aligns with findings among adults and adolescents that perceptions of inequality – though much of this evidence from research on social inequality – especially when viewed as unjustified or insurmountable, predict social-emotional responses, including prosociality, more strongly than objective measures of inequality (e.g., Callan et al., 2017; Du et al., 2024; Du & King, 2022; Goudarzi et al., 2020; Li et al., 2023; Sommet et al., 2026). Our findings suggest that this nuance may emerge quite early in childhood. Future research should further explore the psychological processes underlying this association in development, such as whether perceptions of unfairness reduce children's trust, heighten self-focus, create a scarcity mindset, and/or induce negative emotions that in turn dampen prosocial motivation.

Interestingly, although our manipulations of justified versus unjustified inequality did successfully shift children's average (group-level) fairness perceptions in the predicted ways,

these shifts were not universal. Even in the unjustified disadvantage condition, when children faced the clearest case of unfair inequality, half of them (20 out 40 participants) still viewed it as fair, highlighting the limits of experimental control over children's subjective fairness interpretations. Thus, while our manipulation was intended to test the causal role of perceived unfairness in shaping prosocial behavior, the substantial individual variation in how children interpreted the justifiability of inequality limits strong causal conclusions. The predicted association between unjustified inequality and reduced prosocial behavior emerged primarily among children who subjectively perceived the distribution as unfair, rather than across the full sample as intended.

Why might children differ in their appraisals of the same, even clearly unjustified inequality? One likely explanation is that such appraisals are shaped by broader psychological factors beyond the immediate task, such as prior beliefs about fairness, expectations about how the world works, or developing trust in social systems. Related work in adults has identified similar psychological moderators, including interpersonal trust (Oishi et al., 2011), subjective social status (Rivenbark et al., 2019), and system-justifying ideologies (García-Sánchez et al., 2024). For example, exposure to inequality is associated with reduced trust in others, which in turn lowers expectations of reciprocity and generosity (Lin et al., 2024; Nishi et al., 2015). Our findings align with this pattern. In Study 1, children may have viewed the outcome as accidental or random luck and thus generally maintained trust in the fairness of the system; in contrast, in Study 2, repeated exposure to unjustified disadvantage may have induced an especially strong sense of unfairness among some children and/or more strongly eroded their trust, and therefore reduced their willingness to act

prosocially. Individual differences in trust and system justification, and how sensitive they are to temporary inputs, may stem from early socialization (through parents or school environments) and children's temperaments, and may be difficult to alter via brief experimental manipulations. Children's subjective social status (SSS) may also help explain why some children viewed inequality as more or less fair (Peretz-Lange et al., 2022), although the current work did not measure SSS and thus cannot speak to its role. It will be critical in future work to explore the environmental and psychological mechanisms that shape children's developing interpretations of inequality and, in turn, their (pro)social behaviors.

An alternative account of our findings could be a purely resource-based one: Children who had just received fewer resources were simply less willing to part with additional resources, independent of any social-cognitive appraisals. However, this account cannot fully explain the observed pattern. In Study 2, only children's fairness evaluations – not the objective experience of receiving less – predicted sharing. Furthermore, in an exploratory analysis of the subsample of participants whose fairness judgments aligned with the manipulation, children in Justified-DI and Unjustified-DI – both of whom received fewer resources than the peer – did not both show reduced generosity; rather, only the group that had been unjustifiably disadvantaged did so. In Study 1, moreover, children differentiated social from nonsocial disadvantage in their fairness evaluations, suggesting they were not merely responding to having less but were evaluating the social meaning of the disparity (see also Berelejis et al., 2026). Thus, although simply receiving fewer resources may play some role, it seems unlikely to be the primary mechanism by which resource disparities impact children's prosocial responses.

An important finding in Study 2 was that when the disadvantageous inequality was justified, children between 6 and 9 years perceived it to be as fair as equality (and over the age range, increasingly perceived the two as similar). This finding bolsters previous research showing that as children develop, their third-party fairness evaluations increasingly shift from focusing on the principle of strict equality to other principles, such as merit or need, leading them to find certain forms of inequality acceptable (e.g., Elenbaas & Killen, 2016; Rizzo & Killen, 2016). Importantly, however, our study provides some of the first evidence that by school-age, children can accept some forms of inequality not only as third-party observers but even when it disadvantages them directly, so long as it is justified.

This finding may also offer practical implications for parents and teachers in helping children understand and process experiences of disadvantageous inequality through appropriate justifications, which may in turn buffer children against some of the negative effects of experiencing inequality. However, it is crucial to approach this finding with caution. When using individual merit as a basis for justifying inequality, we may risk overlooking structural or external factors that contribute to differences in merit, such as disparities in access to opportunities and resources (Elenbaas, 2019b) or negative stereotypes about disadvantaged groups (Flanagan, 2013). Moreover, older children and adolescents may perceive societal wealth inequality as relatively fair and as people "getting what they deserve" (Elenbaas & Mistry, 2021). This uncritical acceptance and justification of inequality based on merit may inadvertently perpetuate wealth disparities and may also shape children's motivations and self-concepts. Future research should therefore explore the boundaries of children's justifications for inequality and consider the influences of different social contexts.

The current work also sheds important new light on developing patterns of behavioral versus emotional responses to advantageous and disadvantageous inequality. Study 1 replicated previous findings that both younger and older children perceive disadvantageous inequality as unfair, but only older children view advantageous inequality as unfair (Blake & McAuliffe, 2011; Peretz-Lange et al., 2022). However, children's emotional responses showed a different pattern: While children across the age range reported more negative emotion after experiencing disadvantageous inequality, they reported more positive emotion after experiencing advantageous inequality (or unjustified advantage in Study 2) (see also Gerdemann et al., 2022; Kim et al., 2023; LoBue et al., 2011). This pattern suggests that children's experience of being disadvantaged outweighs considerations of fairness when it comes to their affective responses. Critically, however, we found that it was children's perceptions of fairness rather than their emotional responses that predicted their prosocial behaviors, pointing to the need to disentangle the potentially distinct roles of emotional and cognitive processes in shaping children's reactions to inequality.

Beyond responses to the inequality, children's sympathy for the recipient may operate as a distinct pathway to prosocial behavior. In Study 1, high sympathy for the sick recipient may have masked any effect of resource disadvantage on sharing: When a recipient evokes particularly strong sympathy, it may function as a powerful boundary condition that dominates sharing decisions, leaving little room for the effects of prior inequality. Because sympathy was not measured in Study 2, we cannot rule out that accounting for it would have altered the observed fairness-sharing association. Future research should examine how these pathways interact and under which conditions each takes precedence.

Limitations

Our findings are subject to certain constraints on generalization. First, it remains unclear whether the current findings extend to other forms of prosocial behavior such as helping or cooperation. Resource inequality may be more strongly linked to prosocial behaviors that benefit disadvantaged individuals and rectify inequality, such as sharing or redistribution, than to other prosocial behaviors such as instrumental helping (see Piff & Robinson, 2017). For instance, prior work found that adults from lower social classes were more generous in a dictator game and with charitable donations (Piff et al., 2010), and showed greater support of redistribution (Brown-Iannuzzi et al., 2015), yet they were not more likely to return a misdelivered envelope or to volunteer (Andreoni et al., 2021; Korndörfer et al., 2015).

Second, we measured children's generosity towards an unknown peer outside the immediate inequality context. How experiencing resource disparities may shape prosocial behavior towards specific, known individuals, such as the peer in the competitive game, is an open question. It is possible, for instance, that the competitive context would elicit status anxiety or resentment directed specifically at the opponent, or even at others who have performed better in the same competition, leading to stronger reductions in prosocial behavior towards those individuals than towards unrelated others (cf. Berelejis et al., 2026). Additionally, although inequality in our studies was created by a machine, if it were created by other people, children may show the lowest prosocial motivation (and high punishment motivation) towards the inequality creators (see, e.g., Bernhard et al., 2020; McAuliffe et al., 2015). It is thus important to exercise caution when generalizing findings across different

prosocial behaviors as well as different recipients, given the differing motivations they may involve (Dunfield, 2014; Paulus, 2014, 2018).

We must also consider the difference between long-term exposure to real-world inequality versus brief exposure to resource disparities in an experimental, lab-based setting, which may not result in observable condition effects on children's prosociality. The latter held true in the present work even though we created dyadic interpersonal inequality (rather than a broader environment of inequality) to ensure that children experienced and perceived the disparities firsthand. Yet similar to prior findings that the degree of inequality in children's immediate social environment does not reliably alter sharing (Kirkland et al., 2020, 2021a, 2021b), children's sharing in Study 1 was not altered by the experience of dyadic inequality either. This suggests that a temporary or isolated experience of resource inequality may not, on its own, be sufficient to shape children's prosocial behaviors.

Additionally, prior work with adults shows that perceptions of wealth distribution and justification are associated with other factors such as their SES, cultural views, political attitudes, and so forth (e.g., Brown-Iannuzzi et al., 2015; Kraus et al., 2017; Waldfogel et al., 2021; Wu et al., 2021). These complex relations are increasingly seen among adolescents and children as well (e.g., Elenbaas & Mistry, 2021; Flanagan & Kornbluh, 2019; Wang & Roberts, 2023). Similarly, children's perceptions of inequality in the real world are likely shaped by a complex constellation of factors that brief experimental manipulations cannot fully capture. Still, experimental work is needed as one critical way to identify the possible causal effects and mechanisms underlying the impact of inequality.

Notably, a growing body of adult work suggests that it is not macro-level wealth distributions but more proximal, local-level comparisons – such as comparing oneself with neighbors or peers – that more robustly drive the psychological and behavioral effects of inequality (García-Sánchez et al., 2024; Sánchez-Rodríguez et al., 2025; Willis et al., 2022). Children's daily encounters with resource disparities are likely to take precisely this interpersonal form, suggesting that our dyadic paradigm may capture a psychologically meaningful experience of inequality. Nonetheless, it will be important for future work to examine whether children's fairness evaluations similarly mediate the effects of resource disparities in broader social contexts such as classroom-, neighborhood-, and societal-level inequality, and whether repeated interpersonal experiences of resource disparities shape how children interpret broader structural inequalities over development.

Finally, our sample skewed towards higher-income families. Thus, although in our exploratory analyses, participants' SES was not associated with their prosocial behaviors, this finding may not extend to other SES groups. Prior work has documented a link between SES and prosocial behaviors, albeit with mixed findings (e.g., Andreoni et al., 2021; Chen et al., 2013; Miller et al., 2015; Piff & Robinson, 2017; Safra et al., 2016). Future work should include a more diverse sample and systematically explore the potential moderating effects of children's SES on the relations between inequality and children's social behaviors. Additionally, our U.S.-based sample limits generalizability to other cultures. While recent correlational work on Chinese adolescents' perceptions of societal inequality showed a similar pattern to ours (Li et al., 2023), interpretations of wealth and social status are substantially shaped by culture (Grigoryan et al., 2020; Shu et al., 2024; Wu et al., 2018).

Given the global rise in economic inequality (Chancel et al., 2022), future research should investigate how cultural contexts shape children's responses to both societal and interpersonal inequality.

Conclusion

The present work manipulated 4- to 9-year-old children's experience of interpersonal inequality and suggested that although experiencing disadvantageous inequality impacted children's fairness perceptions and happiness, this impact did not extend to children's sharing with a new peer. Instead, 6- to 9-year-old children were sensitive to the justifiability of disadvantageous inequality, and their perceptions of the fairness of inequality better explained their prosocial behaviors than the objective resource inequality: the more unfair children perceived the inequality to be, the less they shared with the new peer. This work serves as the first systematic examination of the potential role of children's own experience of interpersonal inequality and its perceived fairness on their prosocial behaviors. More broadly, this line of work promises to provide insights into the earliest effects of resource inequality on children's social behavior and to lay the foundation for a novel and important area of research on the near- and long-term impact of inequality on children's social development and wellbeing.

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